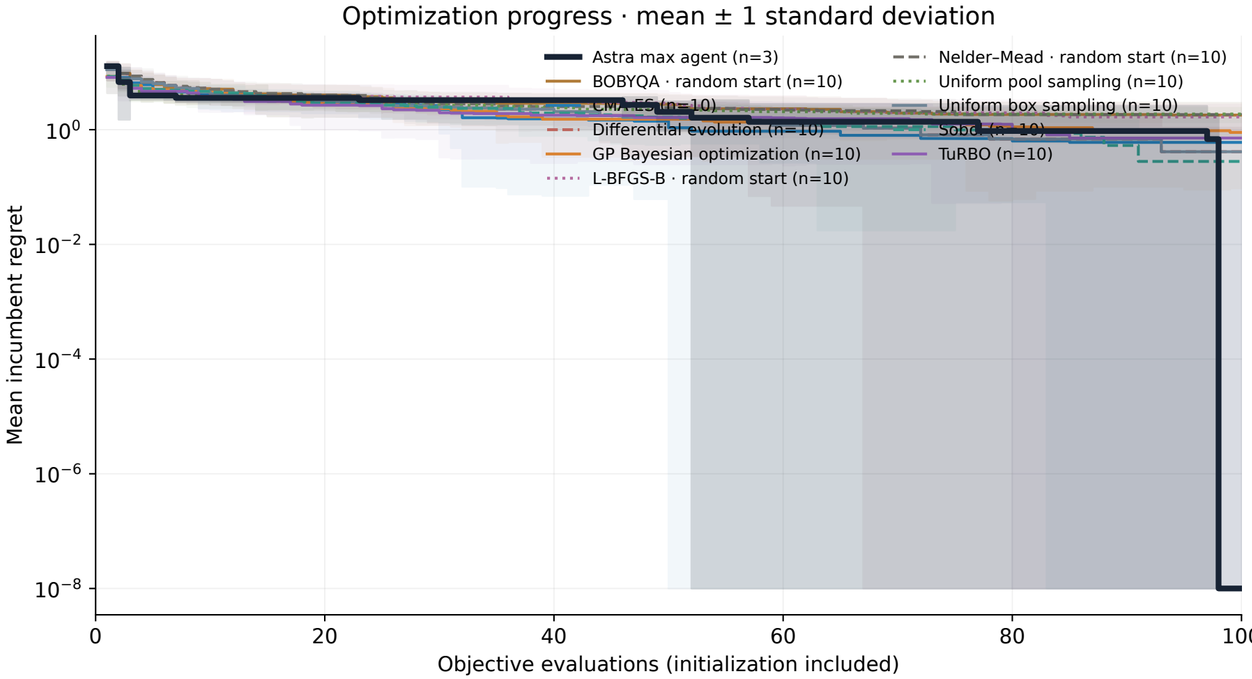
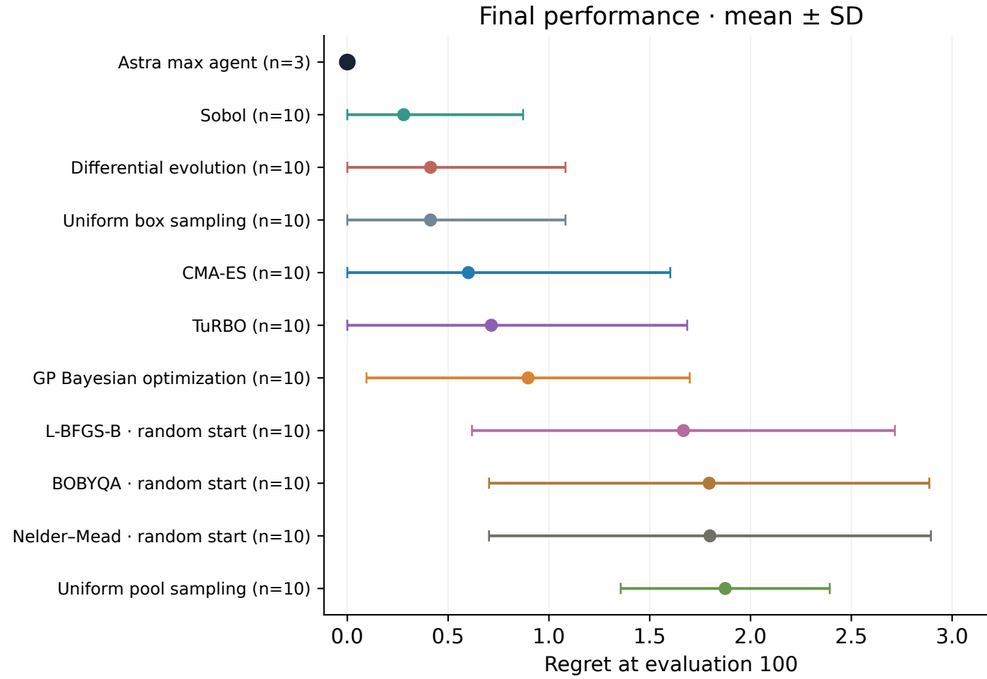
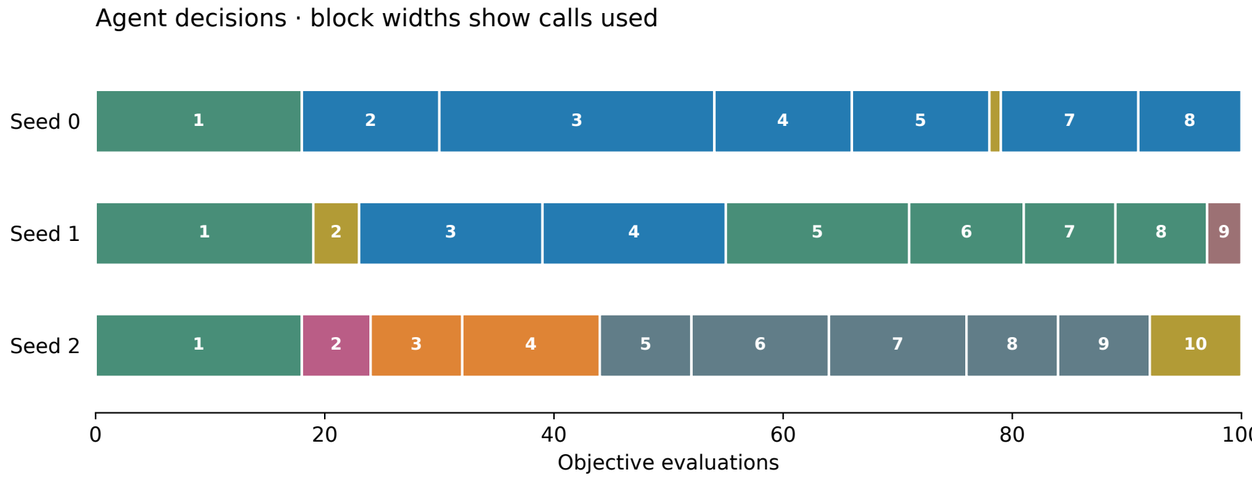
OPV device pool — autonomous agent versus standalone optimizers

526 devices · 11 features · GPT-6 Astra, max reasoning · 3 agent seeds · 100 total evaluations per run







Seed 0
1. Sobol + probes (18) · 2. CMA-ES (12) · 3. CMA-ES (24) · 4. CMA-ES (12) · 5. CMA-ES (12) · 6. Local check (1) · 7. CMA-ES (12) · 8. CMA-ES (9)

Seed 1
1. Sobol + probes (19) · 2. Probes (4) · 3. CMA-ES (16) · 4. CMA-ES (16) · 5. Sobol (16) · 6. Sobol (10) · 7. Sobol (8) · 8. Sobol (8) · 9. Probes (3)

Seed 2
1. Sobol + probes (18) · 2. Probes (6) · 3. GP-BO (8) · 4. GP-BO (12) · 5. GP-BO + Probes (8) · 6. GP-BO + Sobol (12) · 7. GP-BO + Sobol (12) · 8. Revisit + GP-BO (8) · 9. Sobol + Revisit (8) · 10. Revisit (8)

Curves and markers: arithmetic means. Shading/error bars: sample standard deviation (ddof=1), not confidence intervals; lower limits clipped at zero.
Agent n=3; every baseline n=10. BOBYQA, L-BFGS-B and Nelder–Mead use shared seeded uniform starts; original midpoint runs are excluded. Regret = full-pool maximum − best observed score.
Decision boundaries are prospective evaluator snapshots; parentheses give block call counts. Exact labels, settings and rationale are preserved in the per-seed reports.
Log display only: zero values and clipped band endpoints are drawn at 1e-08.